EMS620U-Processing and Analysis in Medical Imaging

VIRTUAL MEDICAL IMAGING LAB REPORT

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INTRODUCTION AND BACKGROUND

Understanding Atrial Fibrillation

Atrial Fibrillation (AF) is a type of arrhythmia or abnormal heartbeat caused by extremely rapid and irregular heartbeats (usually 400 beats per minute) from the atria. A normal heartbeat begins with one electrical impulse from the sinus node in the right atrium (RA) and a healthy person's heart beats around 60 to 150 times per minute. The atria of a person with AF contract irregularly and faster than normal due to faulty electrical signals which cause the atria to get out of sync with the ventricles, which may lead to blood clotting and stroke. Risk factors of AF include old age, alcohol use, obesity, and in some cases can be hereditary. AF is diagnosed through physical examination and definitively through an electrocardiogram, a test which records electrical signals in the heart. AF is the most common heart rhythm abnormality and affects over 33 million people worldwide [1]. Treatment for AF may include stroke assessments, the use of anticoagulants to prevent blood clots, slowing heart rate using beta blockers or calcium channel blockers, restoring and maintaining normal rhythm with antiarrhythmics or cardio diversion [2].

Segmentation Modalities, Challenges and Recent Breakthroughs

Medical imaging modalities such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) have revolutionised the way cardiac structures are visualised and analysed as they generate high-resolution images that enable precise assessment of the atria. Raw imaging data must however undergo segmentation to isolate anatomical regions of interest (ROI). Segmentation involves identifying and extracting ROIs such as tumours and organs from medical images such as CT scans, X-rays and MRI files [3].
These segmentations can be used by doctors to guide treatment planning for AF and other cardiac conditions as they enable accurate targeting of abnormal atrial regions and assist in placing of medical devices such as pacemakers. Segmentations also support risk assessment by helping doctors to analyse atrial morphology, predict complications, and monitor disease progression to evaluate treatment effectiveness over time. Segmentation modalities can be manual, semiautomatic or fully automatic. Some of these methods include Instance segmentation, Semantic segmentation, Panoptic segmentation and Thresholding. Although manual segmentation has been the mainstream method for a long time, it is time-consuming, labour-intensive, and may vary based on the operator. Recent breakthroughs in medical imaging have been driven by advancements in Artificial Intelligence (AI) and deep learning, which have automated the technique, reduced the time and energy needed for segmentation, increased consistency and enabled the analysis of large-scale data. Techniques such as convolutional neural networks (CNNs) and generative adversarial networks (GANs) have improved the precision and efficiency of image segmentation, enabling accurate delineation of complex ROIs like tumours and blood vessels. These have greatly increased the potential for quick detection and diagnosis using medical imaging modalities.

Gaps in Current Research

The task-specific nature of many segmentation models presents a challenge, as their performance often deteriorates when applied to new tasks or datasets. Also, although AI shows great promise in automating segmentation, some gaps in current research include limited generalisability across different imaging methods due to dataset constraints [4], lack of standardised validation metrics leading to inconsistent performance assessments [5], minimal research on integrating manual refinement with AI for improved precision and scalability [6], and significant regulatory and technical barriers hindering clinical translation [7].

Objective of the Study

The main objective of this study is to segment the RA from medical imaging data to analyse its shape and morphology. The 3D Slicer software was employed for manual segmentation of the RA and MATLAB was used to validate the segmentation accuracy against gold-standard data by calculating precision scores and a Dice score to quantify the precision of the segmentation objectively. This study aims to assess the reliability of manual segmentation techniques, explore their implications for clinical applications such as treatment planning and risk assessment in AF, and identify areas for future advancements in segmentation technologies.

METHODOLOGY AND ANALYSES

The clinical data used in this Segmentation was acquired from the publicly available Atrial Segmentation Challenge Data Set made available by the Cardiac Atlas Project. Each 3D MRI patient data was acquired using a clinical whole-body MRI scanner and contained the raw MRI scan and the corresponding ground truth labels for the left atrial (LA) cavity [8]. The software employed to carry out the manual segmentation of the RA on was the 3D slicer (version 5.0.3), an open-source platform used for visualising and processing medical images.

Segmentation Process

- The data was loaded onto the software by uploading the two dataset files, the LA segmentation (laendo.nrrd) and the late gadolinium-enhanced MRI images (lgemri.nrrd), using the "add data" function.
- The "lgemri" file was selected as the primary volume to provide the anatomical reference for segmentation and the "laendo" file containing the segmentation of the LA was loaded as a segmentation mask.
- A new segment named "right atrium" was added in the segment editor module and given a distinct red colour to differentiate it from the LA .
- The intensity thresholds of the scan were then adjusted to capture the intensity ranges corresponding to the RA , which is the primary region of interest.
- The "draw" and "paint" tools were then utilised to precisely map out and delineate the RA slice by slice, as shown in *Figure 1*. The brush sizes were adjusted based on the size of the RA in each slice to not over-segment the RA and the "erase" tool was used to remove over-segmented areas during the segmentation.
- The finalised segmentation was then saved and exported as a nifti file (raendo.nii) for further analysis in MATLAB.

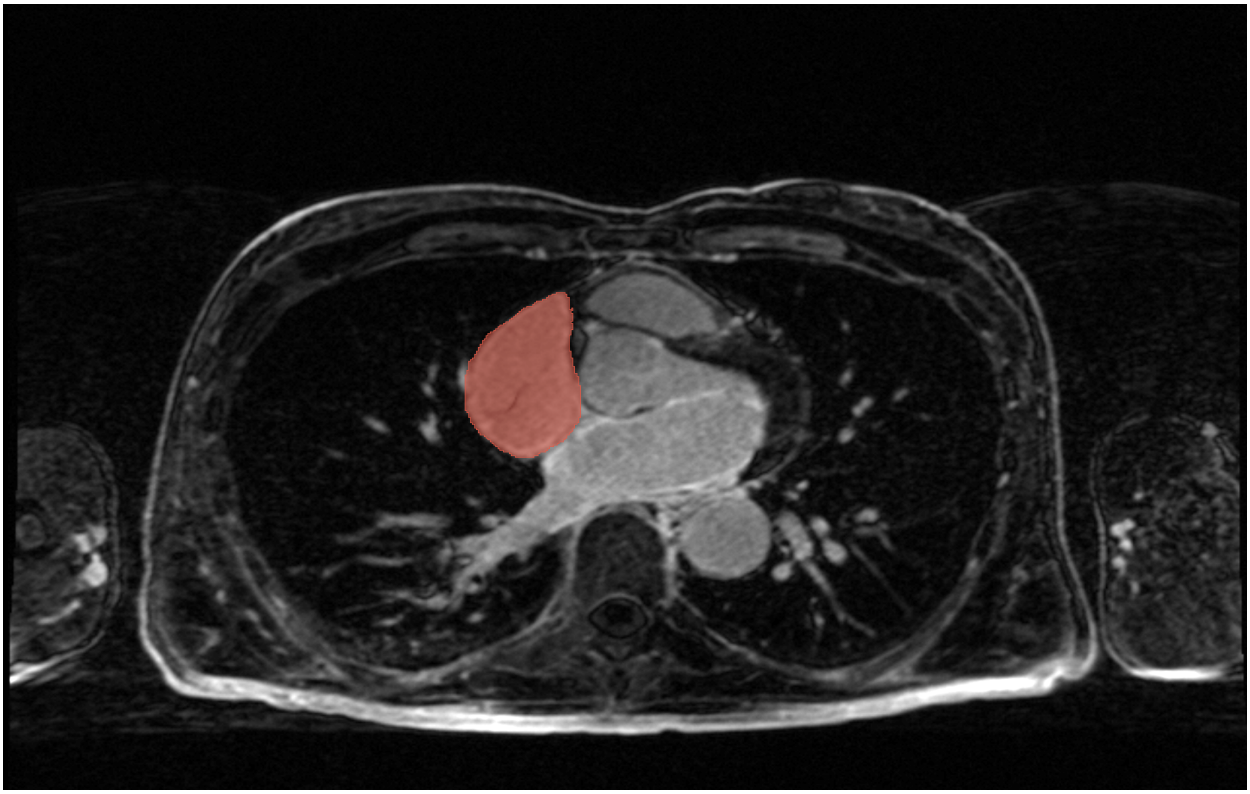


Figure 1: Visual representation of a single slice of the segmented RA on 3D Slicer.

Quantitative Analysis in MATLAB

MATLAB was used to visualise the clinical data, compare the manual RA segmentation with the gold standard segmentation, and to calculate a Dice and Precision Scores to assess the accuracy of the manual segmentation. The analysis was performed using the MATLAB code provided in the appendices, which helped automate these tasks.

The analysis began by preparing the following four files in a folder :

- *lgemri.nii* : The late gadolinium-enhanced MRI scan provided in the dataset, serving as the anatomical reference for evaluating the segmentation quality.
- *laendo.nii* : The pre-segmented LA mask provided in the dataset.
- *raendo.nii :* The manually segmented RA created using 3D slicer.
- *ra_goldstandard.nii* : The gold-standard segmentation for the RA serving as the reference for assessing the accuracy of the manual segmentation.

Loading and Visualising Data in MATLAB as seen in *Appendix A.*

The following steps were used to load and visualise the data in MATLAB, as seen in *Appendices A and B*:

- The four files were imported using the 'niftiread' function, and the working directory was set using the 'cd' command to access the correct files.
- The 'niftiinfo' and 'niftiread' functions were used to load and read the segmentation data for the RA into the variable 'segmentation' which contains the 3D segmentation data of the RA .
- The 'imagesc' function was used to display the 30th slice of the RA segmentation, allowing for a visual assessment of the segmentation for that slice. The 'imagesc' function as seen throughout the MATLAB code in the Appendices was used throughout the analysis to display data as an image that uses the full range of colours in the colourmap.
- The 'lgemri' data was also loaded and read into the variable 'lgemri' and side-by-side plots were generated using the 'imagesc' function to compare the 30th slice of both the segmented RA and the MRI scan to demonstrate alignment, as shown in *Figure 2* below.

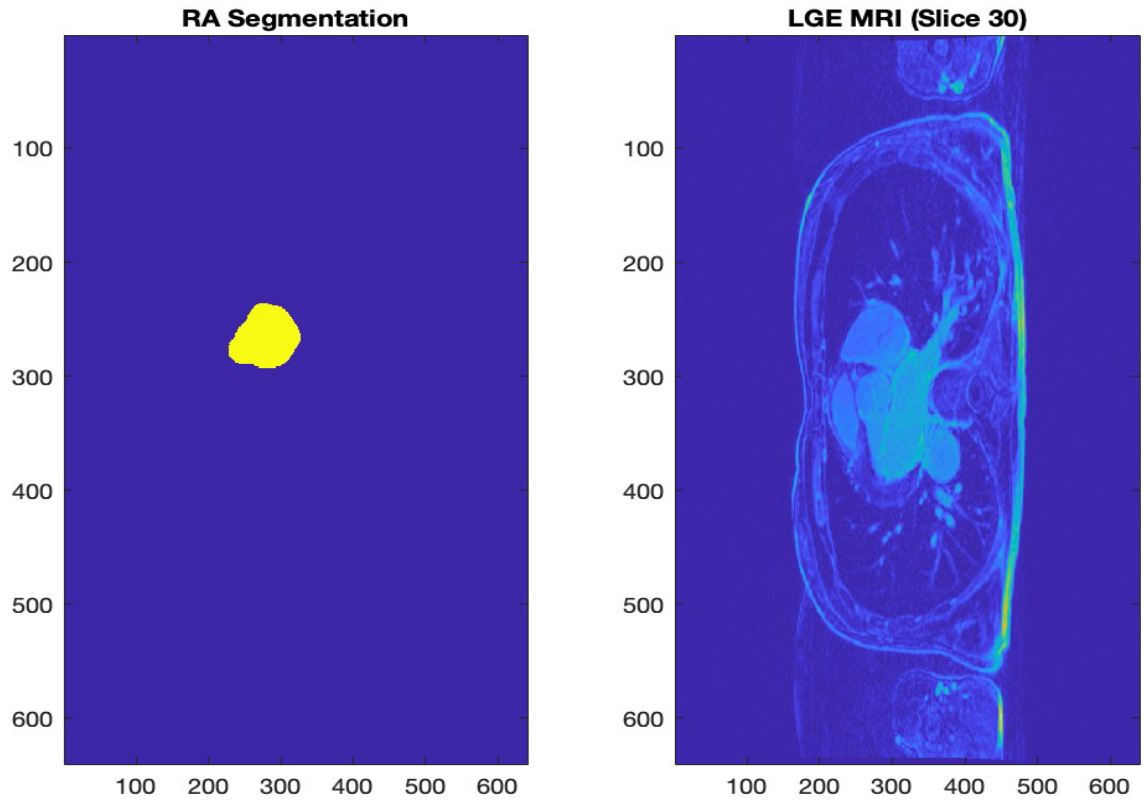


Figure 2: Side-by-side visualisation of the RA segmentation and the MRI scan.

- A ‘loop’ command iterated through all slices of the ‘lgemri’ and RA segmentation data.
- For each slice, side-by-side images of the MRI scan and the segmented RA were displayed and compared.
- The LA segmentation mask and the gold-standard RA segmentation mask were also loaded and read into MATLAB using the ‘niftiread’ function.
- The RA segmentation mask for both the personal RA segmentation and the gold-standard RA segmentation were also combined with the LA segmentation and visualised across all slices to verify the alignment and consistency of both the manually segmented RA and the gold-standard segmentation with the anatomical MRI.
- Similarly, the combined RA gold-standard segmentation and LA segmentation were visualised for all slices, allowing comparison between the manual segmentation and the gold-standard segmentation.
- The selected slice (slice 30) from both the RA segmentation and the gold-standard RA segmentation were then displayed side by side to visually assess how well the manual segmentation aligned with the gold-standard segmentation.
- The combined RA and LA segmentation masks were then saved as new NIfTI files using the ‘niftiwrite’ function to store the results.

Calculation of Precision and Dice Metrics as shown in the MATLAB code in *Appendix B*.

- The 30th slice was selected for both the manual and gold standard segmentation, and the precision score was calculated in for that slice by taking the number of true positives (correctly detected positive pixels) and false positives (background pixels incorrectly detected as positive pixels) into account. An average precision score was then computed across the entire 3D volume, providing a measure of segmentation precision for the entire manual RA segmentation.
- A DICE score was also computed in MATLAB by identifying the number of true positives, false positives, and false negatives (positive overlapping pixels incorrectly detected as background pixels) between the gold standard segmentation and the manual RA segmentation, providing an effective measure of segmentation accuracy.

RESULTS

I-Visual Analysis

Segmentation compared to mask

The manual segmentation of the RA was displayed next to the gold standard mask in MATLAB to be able to visually assess the precision of the manual segmentation. *Figure 3* shows the visual comparison between the manually segmented RA and the gold standard segmentation for the 30th slice. This visual comparison highlights how well the two segmentations are aligned as shown by the axes. It also shows the close agreement between both segmentations in terms of shape and size; however, minor discrepancies are seen in the boundary regions of the manual segmentation. This visual comparison can generally conclude that there is a high degree of similarity between the two segmentations, however this conclusion is subjective as the degree of similarity is not standardised or quantified.

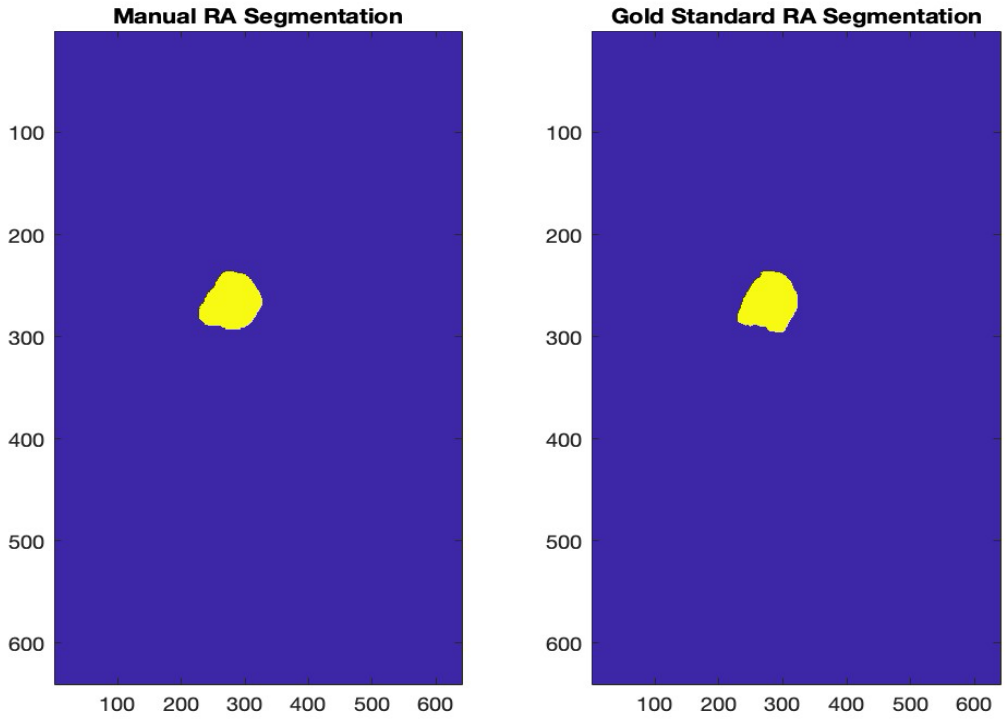


Figure 3: Side-by-side visualization of the manually segmented and gold-standard RA for a selected slice

Combined Segmentation of LA and RA

The combination of the LA and RA segmentations (slice 44) were also displayed next to the "lgemri" scan in MATLAB as shown in *Figure 4*, showing the anatomical relationship between the two atrial structures in the heart. The combined segmentations displayed next to the MRI scan show not only the manual RA segmentation's ability to align with the LA segmentation and revealing the spatial consistency, but also the combined segmentations degree of alignment with the MRI scan. *Figure 4* shows that the two segmentations align well with the MRI scan and there is good spatial consistency between the segmentations of the left and the right atria. However, this analysis is also subject to interpretation by the operator.

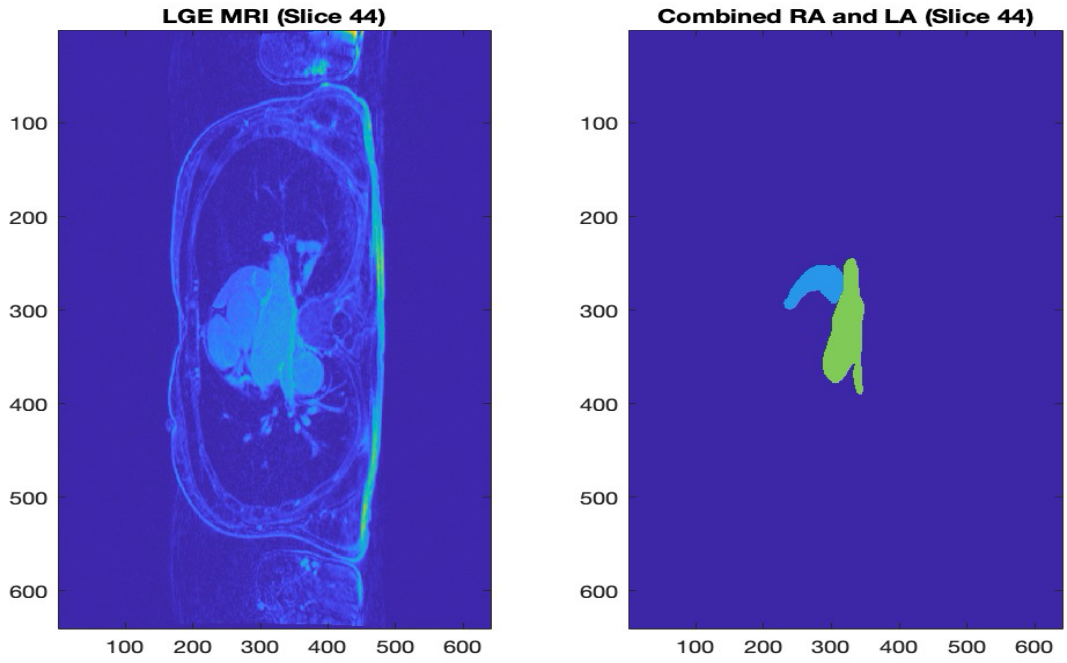


Figure 4: Combined RA (manual) and LA segmentation for an integrated view.

II-Quantitative Analysis

Precision and Recall Scores

The precision score measures the proportion of correctly identified pixels (true positives) to the total number of predicted positives [9]. It is defined as:

$$Precision = \frac{True\ Positives}{True\ Positives + \ False\ Positives}$$

The precision for the 30th slice was calculated as *0.9566* and the overall precision for the entire 3D volume was printed as *0.9165* in the MATLAB output window. This indicates 91.65% of the pixels in the entire segmentation predicted as part of the RA segmentation were correctly classified. Precision scores, however, may be misleading if used independently of other metrics as they focus on avoiding false positives but do not account for missed rue positives. For this reason, it is necessary to introduce Recall, also known as the true positive rate or sensitivity, as it calculates the proportion of true positives to the number of pixels that should have been identified as positive [10]. It is defined as:

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

Using the number of true positives and false negatives printed in the MATLAB code as seen in *Appendix B*, the recall was calculated as *0.4177,* indicating that only 41.77% of the true positive pixels were correctly identified in the ground truth.

Dice Similarity coefficient

The Dice score combines both the Precision and Recall in one metric (using a harmonic mean), providing a balanced assessment of segmentation accuracy [11]. It is defined as:

$$Dice = \frac{2 \times True\ Positives}{2 \times True\ Positives + \ False\ Positives + False\ Negatives}$$

The Dice score for the segmentation was printed in the MATLAB output window as 0.574, indicating 57.7% overlap between the manually segmented RA and the gold standard mask.

DISCUSSION AND CONCLUSIONS

Significance of Quantitative Metrics in Clinical Context

Quantitative metrics provide a standardised and objective way to evaluate segmentation precision in medical imaging. They provide a way to measure how well anatomical boundaries are defined, ensuring that the segmentation accurately represents the underlying anatomy and thus enabling the monitoring of functional and pathological changes. For instance, the size, shape, and volume of cardiac structures can help assess conditions like atrial fibrillation and directly influence clinical decisions. Quantitative metrics allow for the comparing of different segmentation techniques to select the most reliable methods for specific clinical applications. While visual comparison is often used as a preliminary assessment method, it is inherently subjective and prone to variability in interpretation, making it unreliable for evaluating segmentation performance, particularly in clinical applications where precision is key.

Analysis of Segmentation Accuracy

The precision score of *0.9165* for the segmentation was complemented by the Recall Score by capturing the proportion of true positives relative to all ground truth positives. This was done because although the Precision Score is a good metric to work with, it focuses only on minimising false positives but does not account for missed true positives. The recall value of *0.4177* indicates that the model missed identifying 58.23% of the pixels that should have been labelled as positive, emphasizing the need to use both metrics for a more comprehensive evaluation of segmentation accuracy. The Dice coefficient (DSC) is a statistical measure of the similarity between two sets [12]. In medical image segmentation, it quantifies the overlap between a predicted segmentation and a ground truth mask. The value of a DSC ranges from 0, indicating no spatial overlap between two sets of binary segmentation results, to 1, indicating complete overlap [13]. A high DSC value indicates high similarity between the predicted and ground truth masks, meaning that the segmentation model or algorithm is performing well, and a low Dice coefficient value indicates poor segmentation performance [14]. The Dice Similarity Coefficient (DSC) combines precision and recall into a single metric, balancing their trade-offs to provide a comprehensive evaluation of segmentation performance. The DSC is useful in medical imaging, where both false positives and false negatives can significantly impact clinical outcomes. The Dice score of *0.574* calculated for the manual RA segmentation against the gold standard segmentation indicates a moderate level of overlapping between two masks, highlighting that there is room for improvement in the segmentation technique used. A high level of accuracy is important in clinical applications where segmentation outcomes directly impact diagnosis, treatment planning, and risk assessment, especially in conditions like atrial fibrillation ,where accurate segmentations inform treatment ablation therapy and device placement.

Limitations of Approach and Future Directions

The relatively low accuracy as depicted by the dice score obtained by the manual segmentation of the RA could have been as a s result of factors such as under-segmentation of the RA leading to a higher number of false negatives, suboptimal thresholds or human error. Issues such as low image resolution and noise likely contributed to these discrepancies. Employing semi- or fully automated models may be necessary for future work to mitigate these challenges.

Manual segmentations are time-consuming and subject to variability based on the operator, which could affect the reproducibility of the results. For future improvements of segmentation accuracy, it would be sensible to incorporate AI and deep learning models to automate the segmentation. This would help save time, decrease operator dependency and increase the repeatability of the segmentation. Also, other metrics such as Hausdorff distance or Jaccard index could be employed to complement the metrics used in this study to provide a more holistic assessment of segmentation performance.

Additionally, the data obtained from this study could be used to train automated segmentation algorithms such as CNNs by providing labelled manually segmented and gold standard masks to be used to improve segmentation accuracy in clinical applications.

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APPENDIX
PART A

```
clear;clc;close all

%add path
cd("/Users/yemesisa/Library/CloudStorage/OneDrive-
QueenMary,UniversityofLondon/image processing(from wk8)/Segmentation Lab 1411")

%Load the RA nifti file
info=niftiinfo("raendo.nii");
Segmentation=niftiread("raendo.nii");

% Visualise the segmentation slice
figure('Name', 'Segmentation Slice (Slice 30)', 'NumberTitle', 'off');
imagesc(squeeze(Segmentation(:, :, 30)));
title('Segmentation of RA (Slice 30)');

% Add lgemri and compare with segmentation
lgemri=niftiread("lgemri.nii");
figure('Name', 'Segmentation vs LGE MRI (Slice 30)', 'NumberTitle', 'off');
subplot(1,2,1);
imagesc(squeeze(Segmentation(:, :, 30)));
title('RA Segmentation');
subplot(1,2,2);
imagesc(squeeze(lgemri(:, :, 30)));
title('LGE MRI (Slice 30)');

% Looping over all slices of the file
for index=1:size(lgemri, 3)
figure(100)
subplot(1, 2,1)
imagesc(squeeze(lgemri(:, :, index)))
subplot(1,2,2)
imagesc(squeeze(Segmentation(:, :, index)))
pause(0.1)
end

% Read in LA .nii file
LA_segmentation=niftiread("laendo.nii");

% REading in gold standard segmentation
Gold_standard=niftiread("ra_goldstandard.nii");

% Combine both LA segmentation and RA segmentation (gold standard or your
% RA personal one)
Both_Atria=Segmentation+2.*LA_segmentation; % for personal one
Both_Atria_goldstandard=Gold_standard+2.*LA_segmentation; % for gold standard one

%Visualise it together (RA + LA)
figure('Name', 'Combined RA and LA Segmentation', 'NumberTitle', 'off');
for index = 1:size(lgemri, 3)
```

PART A continued

```
    subplot(1, 2, 1);
    imagesc(squeeze(lgemri(:, :, index)));
    title(sprintf('LGE MRI (Slice %d)', index));
    subplot(1, 2, 2);
    imagesc(squeeze(Both_Atria(:, :, index)));
    title(sprintf('Combined RA and LA (Slice %d)', index));
    pause(0.1);
end

%Visualise it together (RA gold standard + LA)
figure('Name', 'RA Gold Standard vs Combined LA', 'NumberTitle', 'off');
for index = 1:size(lgemri, 3)
    subplot(1, 2, 1);
    imagesc(squeeze(lgemri(:, :, index)));
    title(sprintf('LGE MRI (Slice %d)', index));
    subplot(1, 2, 2);
    imagesc(squeeze(Both_Atria_goldstandard(:, :, index)));
    title(sprintf('Gold Standard RA and LA (Slice %d)', index));
    pause(0.1);
end


% The following lines of code will save the  nifti will write the
% combination of RA and LA together from the previous two plots

% Write the combinaiton of both into .nii file (for RA + LA)
niftiwrite(Both_Atria, 'Both_atria.nii')

% Write the combinaiton of both into .nii file (for RA gold standard + LA)
niftiwrite(Both_Atria_goldstandard, 'Both_atria_goldstandard.nii')
```

PART B

```
% Calculating Precision Score
% we already have our RA nifti file read in so no need to re read it again
% Our RA nifti file is the Segmentation variable
% Our RA gold standard is the Gold_Gold_standard variable

%But... as a refresh here is the code for reading them in:
% Change to your working directory
cd("/Users/yemesisa/Library/CloudStorage/OneDrive-
QueenMary,UniversityofLondon/image processing(from wk8)/Segmentation Lab 1411");

% Read NIfTI files
Segmentation = niftiread('raendo.nii');
Gold_standard = niftiread('ra_goldstandard.nii');

% We then pick a slice where we want to calculate the accuracy between
% our RA segmentation and our RA gold standard
% Select a slice for comparison
sliceno = 30; % Adjust as needed

% Extract the corresponding slices
```

```
Slice_RA = Segmentation(:, :, sliceno); % RA segmentation
Slice_gold_standard = Gold_standard(:, :, sliceno); % Gold standard

% Adding our values to our true positives/false positives to calculate
% precision for an individual slice
true_positives=sum((Slice_gold_standard(:)==1) & (Slice_RA(:)==1));
false_positives=sum((Slice_gold_standard(:)==0) & (Slice_RA(:)==1));

precision_oneslice=true_positives/(true_positives+false_positives);

fprintf('Precision for one specified slice: %.4f\n', precision_oneslice)

% Visualising the slices between our RA segmentation and gold standard
figure('Name', 'GS vs RA', 'NumberTitle', 'off') % Set figure name and turn off
numbering
subplot(1,2,1)
imagesc(Slice_RA)
title('Manual RA Segmentation')
subplot(1,2,2)
imagesc(Slice_gold_standard)
title('Gold Standard RA Segmentation')

% Calculating precision score between our entire RA segmentation and gold standard
true_positives=sum((Gold_standard(:)==1) & (Segmentation(:)==1));
false_positives=sum((Gold_standard(:)==0) & (Segmentation(:)==1));

precision_allslice=true_positives/(true_positives+false_positives);

fprintf('Precision for entire segmentation: %.4f\n', precision_allslice)
```

PART B continued

```
% Load or define groundtruth_mask and pred_mask here
% Replace with the actual data or load from files
groundtruth_mask = niftiread('ra_goldstandard.nii'); % Example ground truth mask
pred_mask = niftiread('raendo.nii'); % Example predicted mask

% Call the function and calculate the Dice score
dice_score = dice_coef(groundtruth_mask, pred_mask);

% Print a detailed message
fprintf('The Dice coefficient for the segmentation is %.3f.\n', dice_score);


function dice = dice_coef(groundtruth_mask, pred_mask)
    % Calculate True Positives (TP)
    TP = sum((groundtruth_mask(:) == 1) & (pred_mask(:) == 1));

    % Calculate False Positives (FP)
    FP = sum((groundtruth_mask(:) == 0) & (pred_mask(:) == 1));

    % Calculate False Negatives (FN)
```

```
    FN = sum((groundtruth_mask(:) == 1) & (pred_mask(:) == 0));

    % Compute the Dice coefficient
    dice = (2 * TP) / (2 * TP + FP + FN);

    % Round the Dice coefficient to 3 decimal places
    dice = round(dice, 3);

    % Print the results
    fprintf('True Positives (TP): %d\n', TP);
    fprintf('False Positives (FP): %d\n', FP);
    fprintf('False Negatives (FN): %d\n', FN);
    fprintf('Dice Score: %.3f\n', dice);
end
```